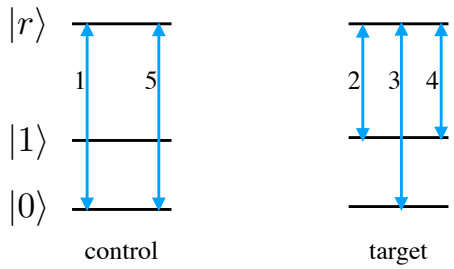
Question 1.

A Rydberg gate is shown below for two adjacent atoms to be entangled. Each of the five pulses composing the sequence are π pulses.



Part A (grade: 1-5)

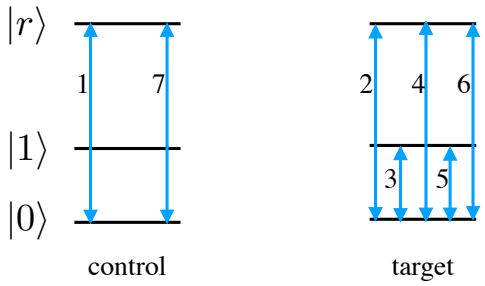
1. Show the effect on each of the qubit basis states $|0_c 0_t\rangle$, $|0_c 1_t\rangle$, $|1_c 0_t\rangle$, $|1_c 1_t\rangle$ from applying the pulse sequence. Explain your reasoning.

2. Write out the 4 ×4 matrix resulting from the pulse sequence.

3. In comparison to the scheme described in class describe a technical disadvantage to this approach to realising a CNOT equivalent gate.

Part B (Advanced: grade +0,+1,+2)

An additional two π pulses applied to the target during the gate sequence can be used to realise a similar version of the CNOT gate, by introducing additional rotations to the target qubit. This results in a seven-pulse sequence.



a) Determine the resulting action of the gate on the $|0_c 0_t\rangle$, $|0_c 1_t\rangle$, $|1_c 0_t\rangle$, $|1_c 1_t\rangle$ basis.

b) Determine a circuit that will result in the canonical CNOT gate via additional single qubit rotations.